

```
# 3/3/26
# General pipeline for ChIP-seq, CUT&RUN/TAG, DRIP-seq, DRIPc-seq(SE & PE) fastq --
> bigwig (for IGV)
```

```
#####
```

```
# USER VARIABLES: Change as needed
```

```
#####
```

```
MARK="TDP43"           # H3K9me3, R-loop, TDP-43, G4
ASSAY_TYPE="CHIP"      # CHIP, CNR, DRIP, DRIPc
GSE="GSE91745"
CELL_TYPE="Bcell"
READ_MODE="PE"         # SE for single-end, PE for paired-end
```

```
WORKDIR="/home/$USER/scr4_hlee308/$USER/datasets/${MARK}/${ASSAY_TYPE}/${GSE}/${CELL_TYPE}"
```

```
SRR_IDS=("SRR5111562")
INPUT_IDS=("SRR5111556")      # For DRIPc, leave empty
```

```
ALL_IDS=("${SRR_IDS[@]}" "${INPUT_IDS[@]}")
GENOME="hg38"
FASTQDIR="$WORKDIR/fastq"
TRIMDIR="$WORKDIR/trimmedreads"
SAMDIR="$WORKDIR/samfiles"
BAMDIR="$WORKDIR/bamfiles"
MACS3DIR="$WORKDIR/macs3peaks"
BIGWIGDIR="$WORKDIR/bigwigfiles"
BOWTIE2_INDEX="/home/$USER/scr4_hlee308/$USER/refgenome/BOWTIE2refgenome/hg38refgenome/hg38"
```

```
cd $WORKDIR
```

```
# Make WORKDIR yourself
mkdir -p "$WORKDIR"/{fastq,trimmedreads,samfiles,bamfiles,macs3peaks,bigwigfiles}
```

```
cd $WORKDIR
```

```
#####  
# Download FASTQ files (10min)  
#####
```

```
# Make the download fastq script  
cd $WORKDIR  
nano download_fastq.slurm
```

```
# PASTE BELOW:
```

```
#####  
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#####  
#!/bin/bash  
#SBATCH --job-name=download_fastq  
#SBATCH --output=download_fastq_%j.out  
#SBATCH --nodes=1  
#SBATCH --ntasks=1  
#SBATCH --cpus-per-task=4  
#SBATCH --time=00:20:00  
#SBATCH --partition=shared  
#SBATCH -A hlee308  
#SBATCH --mail-type=END,FAIL  
#SBATCH --mail-user=dsaghai1@jh.edu
```

```
echo "START JOB: $(date)"
```

```
# ----- Load modules -----  
module load sra-tools/3.0.3
```

```
# ----- Define variables -----  
SRR=$1  
FASTQDIR=$2  
READ_MODE=$3
```

```
cd $FASTQDIR
```

```
# ----- Download all SRRs -----  
prefetch "$SRR"
```

```
if [[ "$READ_MODE" == "SE" ]]; then  
    fasterq-dump -e $SLURM_CPUS_PER_TASK "$SRR"  
else  
    fasterq-dump --split-files -e $SLURM_CPUS_PER_TASK "$SRR"  
fi
```

```
# ----- Delete SRR folders -----  
find . -maxdepth 1 -type d -name 'SRR*' -exec rm -rf {} +
```

```
echo "END JOB: $(date)"
```

```
# OUTPUT  
# in FASTQDIR  
# single-end: SRRXXXXXX.fastq  
# paired-end: SRRXXXXXX_1.fastq, SRRXXXXXX_2.fastq
```

```
#####  
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```

```
# Run the script  
cd $WORKDIR  
module load slurm
```

```
for SRR in "${ALL_IDS[@]}"; do  
    sbatch download_fastq.slurm "$SRR" "$FASTQDIR" "$READ_MODE"  
done
```

```
mv download*.out $FASTQDIR
```

```
# OUTPUT  
# in FASTQDIR  
# single-end: SRRXXXXXX.fastq  
# paired-end: SRRXXXXXX_1.fastq, SRRXXXXXX_2.fastq
```

```
#####  
# TrimGalore: Trim adapters & low quality bases on fastq file (10min)  
#####
```

```
# Make the trimgalore script  
cd $WORKDIR  
nano trimgalore.slurm
```

PASTE BELOW:

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```

#!/bin/bash

#SBATCH --job-name=trimgalore

#SBATCH --output=trimgalore_%j.out

#SBATCH --nodes=1

#SBATCH --ntasks=1

#SBATCH --cpus-per-task=8

#SBATCH --time=01:00:00

#SBATCH --partition=shared

#SBATCH -A hlee308

#SBATCH --mail-type=END,FAIL

#SBATCH --mail-user=dsaghai1@jh.edu

echo "START JOB: \$(date)"

----- Load conda to load trimgalore_env -----

source /data/apps/extern/spack_on/gcc/9.3.0/

anaconda3/2024.02-1-7cjokusndbeun4sajttdsp5jhe6jqsmmt/etc/profile.d/conda.sh

conda activate trimgalore_env

----- Load modules -----

module load fastqc/0.11.9

----- Define variables -----

SRR=\$1

FASTQDIR=\$2

TRIMDIR=\$3 # output folder for trimmed fastq

READ_MODE=\$4

```

# Run trimgalore
if [[ "$READ_MODE" == "SE" ]]; then
    trim_galore --cores $SLURM_CPUS_PER_TASK --fastqc --gzip --length 20 -q
    20 --output_dir $TRIMDIR $FASTQDIR/${SRR}.fastq

else
    trim_galore --paired --cores $SLURM_CPUS_PER_TASK --fastqc --gzip --
length 20 -q 20 --output_dir $TRIMDIR $FASTQDIR/${SRR}_1.fastq $FASTQDIR/${SRR}
_2.fastq
fi

```

```

echo "END JOB: $(date)"

```

```

# OUTPUT:

```

```

# in TRIMDIR
# single-end: ${SRR}_trimmed.fq.gz
# paired-end: ${SRR}_1_val_1.fq.gz ${SRR}_2_val_2.fq.gz

```

```

# Notes:

```

```

# --fastqc: makes FASTQC report after trimming
# --gzip: compresses output FASTQ files
# --length 20: throw away trimmed reads shorter than 20bp (default)
# --q 20: (default) trim low-quality ends (<Q20)
# --output_dir: trimmed FASTQ files are located in TRIMDIR

```

```

#####
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```

module load slurm

```

```

for SRR in "${ALL_IDS[@]}"; do
    sbatch trimgalore.slurm "$SRR" "$FASTQDIR" "$TRIMDIR" "$READ_MODE"
done

```

```
mv *trimgalore*.out "$TRIMDIR"
```

```
rm -rf "$FASTQDIR"
```

```
# OUTPUT:
```

```
    # in TRIMDIR
```

```
    # single-end: ${SRR}_trimmed.fq.gz
```

```
    # paired-end: ${SRR}_1_val_1.fq.gz ${SRR}_2_val_2.fq.gz
```

```
#####
```

```
# Bowtie2: Align sequence to reference genome (3hr) (Henikoff parameters)
```

```
# Samtools: sort, index BAM (Malachowski et al., 2023 parameters, but without  
downsampling)
```

```
#####
```

```
# 2/27/26: removed --no-mixed --no-discordant for chip PE alignment (those were  
Malachowski 2023 PE parameters), since % aligned was much lower than 90%
```

```
# Make the bowtie2 samtools script  
nano bowtie2_samtools_bam.slurm
```

```
# PASTE BELOW:
```

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```

```
#!/bin/bash  
#SBATCH --job-name=bowtie2_samtools_bam  
#SBATCH --output=bowtie2_samtools_bam_%j.out  
#SBATCH --nodes=1  
#SBATCH --ntasks=1  
#SBATCH --cpus-per-task=4  
#SBATCH --time=05:00:00  
#SBATCH --mem=48G  
#SBATCH --partition=shared  
#SBATCH -A hlee308  
#SBATCH --mail-type=END,FAIL  
#SBATCH --mail-user=dsaghai1@jh.edu
```

```
set -euo pipefail
```

```
echo "START JOB: $(date)"
```

```
# ----- Load tools -----  
source /data/apps/extern/spack_on/gcc/9.3.0/  
anaconda3/2024.02-1-7cjzusndbeun4sajttdsp5jhe6jqsmnt/etc/profile.d/conda.sh  
conda activate bowtie2_env  
module load samtools
```

```
# ----- Define variables -----  
SRR=$1  
TRIMDIR=$2  
BAMDIR=$3  
BOWTIE2_INDEX=$4  
READ_MODE=$5  
ASSAY_TYPE=$6
```

```
echo "ALIGNING SRR: $SRR"
```



```

# Temporary files
NAMESORTED_BAM="${BAMDIR}/${SRR}.namesorted.bam"
FIXMATE_BAM="${BAMDIR}/${SRR}.fixmate.bam"
COORDSORT_BAM="${BAMDIR}/${SRR}.sorted.bam"
RMDUP_BAM="${BAMDIR}/${SRR}.rmdup.bam"
FINAL_BAM="${BAMDIR}/${SRR}.final.bam"

#####
# BOWTIE2
#####

# ----- Bowtie2 alignment piped directly to name-sorted BAM -----
# DRIP & DRIPc use DEFAULT bowtie2 parameters for SE and PE
if [[ "$ASSAY_TYPE" == "DRIP" || "$ASSAY_TYPE" == "DRIPc" ]]; then
    if [[ "$READ_MODE" == "SE" ]]; then
        bowtie2 \
            --no-unal \
            -x "$BOWTIE2_INDEX" \
            -U "$TRIMDIR/${SRR}_trimmed.fq.gz" \
            -p "$SLURM_CPUS_PER_TASK" \
            | samtools view -@ "$SLURM_CPUS_PER_TASK" -b - \
            | samtools sort -n -@ "$SLURM_CPUS_PER_TASK" -o "$NAMESORTED_BAM" -
    else
        bowtie2 \
            --no-unal \
            -x "$BOWTIE2_INDEX" \
            -1 "$TRIMDIR/${SRR}_1_val_1.fq.gz" \
            -2 "$TRIMDIR/${SRR}_2_val_2.fq.gz" \
            -p "$SLURM_CPUS_PER_TASK" \
            | samtools view -@ "$SLURM_CPUS_PER_TASK" -b - \
            | samtools sort -n -@ "$SLURM_CPUS_PER_TASK" -o "$NAMESORTED_BAM" -
    fi
fi

# CUT&RUN & CHIP use HENIKOFF alignment parameters
else

```

```

if [[ "$READ_MODE" == "SE" ]]; then
    bowtie2 \
        --no-unal \
        -x "$BOWTIE2_INDEX" \
        -U "$TRIMDIR/${SRR}_trimmed.fq.gz" \
        -p "$SLURM_CPUS_PER_TASK" \
        | samtools view -@ "$SLURM_CPUS_PER_TASK" -b - \
        | samtools sort -n -@ "$SLURM_CPUS_PER_TASK" -o "$NAMESORTED_BAM" -
else
    bowtie2 \
        --no-unal \
        --local \
        --very-sensitive-local \
        -I 10 -X 700 \
        -x "$BOWTIE2_INDEX" \
        -1 "$TRIMDIR/${SRR}_1_val_1.fq.gz" \
        -2 "$TRIMDIR/${SRR}_2_val_2.fq.gz" \
        -p "$SLURM_CPUS_PER_TASK" \
        | samtools view -@ "$SLURM_CPUS_PER_TASK" -b - \
        | samtools sort -n -@ "$SLURM_CPUS_PER_TASK" -o "$NAMESORTED_BAM" -
fi
fi

```

BOWTIE2 NOTES

```

# --local: allows partial alignment
# --very-sensitive-local: catches harder alignments
# --no-unal: don't output unmapped reads into SAM
# --no-mixed: (PE ONLY) don't output alignments where only 1 mate maps
# --no-discordant: (PE ONLY) mates must align w correct orientation & distance
# -I 10 -X 700: (PE ONLY) 10 bp min expected insert size; 700 bp max expected
insert size.

```

DRIP & DRIPc → default Bowtie2 (SE + PE) (from Chedin)

ChIP & CUT&RUN → default Bowtie2

```

#####
# SAMTOOLS
#####

# ----- fixmate (PE only) -----
if [[ "$READ_MODE" == "PE" ]]; then
    samtools fixmate -m "$NAMESORTED_BAM" "$FIXMATE_BAM"
else
    cp "$NAMESORTED_BAM" "$FIXMATE_BAM"
fi

# ----- Coordinate sort -----
samtools sort -@ "$SLURM_CPUS_PER_TASK" -o "$COORDSORT_BAM"
"$FIXMATE_BAM"

# ----- Remove duplicates -----
samtools markdup -r -@ "$SLURM_CPUS_PER_TASK" "$COORDSORT_BAM"
"$RMDUP_BAM"

# ----- Remove unmapped reads -----
samtools view -@ "$SLURM_CPUS_PER_TASK" -b -F 4 "$RMDUP_BAM" >
"$FINAL_BAM"
samtools index -@ "$SLURM_CPUS_PER_TASK" "$FINAL_BAM"

# ----- DRIPc-only strand separation -----
if [[ "$ASSAY_TYPE" == "DRIPc" ]]; then
    echo "Separating DRIPc reads by strand..."

    if [[ "$READ_MODE" == "SE" ]]; then
        # SE DRIPc
        samtools view -@ "$SLURM_CPUS_PER_TASK" -b -f 16 -F 256 "$FINAL_BAM" >
"$BAMDIR/${SRR}.minus.bam"
        samtools view -@ "$SLURM_CPUS_PER_TASK" -b -F 16 -F 256 "$FINAL_BAM" >
"$BAMDIR/${SRR}.plus.bam"
        samtools index -@ "$SLURM_CPUS_PER_TASK" "$BAMDIR/${SRR}.plus.bam"
        samtools index -@ "$SLURM_CPUS_PER_TASK" "$BAMDIR/${SRR}.minus.bam"
    else
        # PE DRIPc (READ 1 only)

```

```

        samtools view -@ "$SLURM_CPUS_PER_TASK" -b -f 80 -F 256 "$FINAL_BAM" >
"$BAMDIR/${SRR}.R1.minus.bam"
        samtools view -@ "$SLURM_CPUS_PER_TASK" -b -f 64 -F 272 "$FINAL_BAM" >
"$BAMDIR/${SRR}.R1.plus.bam"
        samtools index -@ "$SLURM_CPUS_PER_TASK" "$BAMDIR/${SRR}.R1.plus.bam"
        samtools index -@ "$SLURM_CPUS_PER_TASK" "$BAMDIR/${SRR}.R1.minus.bam"
    fi
fi

```

```

# ----- Cleanup intermediates -----
rm -f "$NAMESORTED_BAM" "$FIXMATE_BAM" "$COORDSORT_BAM"
# Keep rmdup for QC/debug; uncomment if you want to remove it:
# rm -f "$RMDUP_BAM"

echo "END JOB: $(date)"

```

```

# OUTPUT (per SRR):
#   in $BAMDIR
#   ${SRR}.final.bam           # mapped, coordinate-sorted, deduplicated BAM
#   ${SRR}.final.bam.bai       # BAM index
#   ${SRR}.rmdup.bam           # deduplicated BAM before final -F 4 filter (kept for QC/
debug)
#
#   DRIPc only:
#   SE:
#     ${SRR}.plus.bam
#     ${SRR}.plus.bam.bai
#     ${SRR}.minus.bam
#     ${SRR}.minus.bam.bai
#   PE:
#     ${SRR}.R1.plus.bam
#     ${SRR}.R1.plus.bam.bai
#     ${SRR}.R1.minus.bam

```

```

#   ${SRR}.R1.minus.bam.bai
#
#   in submit directory (then moved if you run mv command):
#   bowtie2_samtools_bam_<jobid>.out  # contains START/END timestamps + Bowtie2
alignment summary

#####
#####
#####
#####
module load slurm

BOWTIE2_INDEX="/home/$USER/scr4_hlee308/$USER/refgenome/BOWTIE2refgenome/
hg38refgenome/hg38"

for SRR in "${ALL_IDS[@]}"; do
    sbatch bowtie2_samtools_bam.slurm "$SRR" "$TRIMDIR" "$BAMDIR"
"$BOWTIE2_INDEX" "$READ_MODE" "$ASSAY_TYPE"
done

mv *bowtie2_samtools_bam*.out "$BAMDIR"

```

```

#####
# MACS3 (still WIP)
# Default MACS3 parameters
#####
# Make the macs3_peakcall script
nano macs3_peakcall.slurm

```

PASTE BELOW:

```
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#####
```

#!/bin/bash

#SBATCH --job-name=macs3_peakcall

#SBATCH --output=macs3_peakcall_%j.out

#SBATCH --nodes=1

#SBATCH --ntasks=1

#SBATCH --cpus-per-task=1

#SBATCH --time=00:20:00

#SBATCH --partition=shared

#SBATCH -A hlee308

#SBATCH --mail-type=END,FAIL

#SBATCH --mail-user=dsaghai1@jh.edu

set -euo pipefail

echo "START JOB: \$(date)"

module load MACS/3.0.0

TEST=\$1

INPUT=\$2 # pass "NONE" for DRIP/DRIPc without control

BAMDIR=\$3

MACS3DIR=\$4

ASSAY_TYPE=\$5

READ_MODE=\$6

GSE=\$7

GENOME=2913022398

TEST_BAM="\${BAMDIR}/\${TEST}.final.bam"

if [["\$INPUT" == "NONE" || -z "\$INPUT"]]; then

if [["\$READ_MODE" == "SE"]]; then

macs3 callpeak -t "\$TEST_BAM" -f BAM -g "\$GENOME" -n "\${GSE}_\${ASSAY_TYPE}_\${TEST}" -B -q 0.01 --outdir "\$MACS3DIR"

else

macs3 callpeak -t "\$TEST_BAM" -f BAMPE -g "\$GENOME" -n "\${GSE}_\${ASSAY_TYPE}_\${TEST}" -B -q 0.01 --outdir "\$MACS3DIR"

```

fi
else
  INPUT_BAM="${BAMDIR}/${INPUT}.final.bam"
  if [[ "$READ_MODE" == "SE" ]]; then
    macs3 callpeak -t "$TEST_BAM" -c "$INPUT_BAM" -f BAM -g "$GENOME" -n "${GSE}_${ASSAY_TYPE}_${TEST}" -B -q 0.01 --outdir "$MACS3DIR"
  else
    macs3 callpeak -t "$TEST_BAM" -c "$INPUT_BAM" -f BAMPE -g "$GENOME" -n "${GSE}_${ASSAY_TYPE}_${TEST}" -B -q 0.01 --outdir "$MACS3DIR"
  fi
fi
fi

```

```

#add --broad for h3k9me3 and broad peaks

```

```

echo "END JOB: $(date)"

```

```

# OUTPUT

```

```

  #Peak files
  #PREFIX_peaks.narrowPeak (peak calls; TF-style)
  #PREFIX_peaks.xls       (tabular peak stats)
  #PREFIX_summits.bed     (summit position per peak)
  #Signal tracks (bedGraph) from -B
  #PREFIX_treat_pileup.bdg (treatment pileup signal)
  #PREFIX_control_lambda.bdg (local background model; from control)
#####
#####
#####
#####

```

```

# Run bgwgcompare_bigwig.slurm
module load slurm

```

```

INPUT="${INPUT_IDS[0]}"
# For DRIP/DRIPc, INPUT="NONE"

```

```
for TEST in "${SRR_IDS[@]}"; do
    sbatch macs3_peakcall.slurm \
        "$TEST" "$INPUT" "$BAMDIR" "$MACS3DIR" "$ASSAY_TYPE" "$READ_MODE"
"$GSE"
done
```

```
mv macs3_peakcall*.out "$MACS3DIR"
```

```
# output:
```

```
#####
# Deeptools: Convert sorted BAM file to BigWig file and subtract/ratio the input signal
(>30min)
# Malachowski et al. 2023 paramters, but without samflag
#####
```

```
# Make the bigwigcompare script
nano bgwgcompare_bigwig.slurm
```

```
# PASTE BELOW:
#####
#####
#####
#####
#!/bin/bash
```



```
#SBATCH --job-name=bgwgcompare_bigwig
#SBATCH --output=bgwgcompare_bigwig_%j.out
#SBATCH --nodes=1
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=4
#SBATCH --time=02:00:00
#SBATCH --mem=32G
#SBATCH --partition=shared
#SBATCH -A hlee308
#SBATCH --mail-type=END,FAIL
#SBATCH --mail-user=dsaghai1@jh.edu
```

```
echo "START JOB: $(date)"
```

```
# ----- Load conda -----
source /data/apps/extern/spack_on/gcc/9.3.0/
anaconda3/2024.02-1-7cjozusndbeun4sajttdsp5jhe6jqsmnt/etc/profile.d/conda.sh
conda activate /home/dsaghai1/.conda/envs/deeptools_env
```

```
# ----- Define variables -----
TEST=$1      # ChIP / DRIP / CUT&RUN sample, no suffix
INPUT=$2     # input / IgG / control, no suffix
BAMDIR=$3    # dir with *.rmdup.bam or *.markdup.bam
BIGWIGDIR=$4 # output dir for bigWigs
ASSAY_TYPE=$5 # CHIP, CUTNRUN, CUTNTAG, DRIP, DRIPc
READ_MODE=$6 # SE or PE
FRAGLEN=$7   # 0 = PAIRED-END, >0 = SINGLE-END fragment length from MACS3
MARK=$8      # SOX2, G4, H3K9me3, R-loop, TDP-43
```

```
GENOME=hg38
EFFECTIVE_GENOME_SIZE=2913022398 # hg38 only
BINSIZE=10
SMOOTH=30
```

```
BAM_SUFFIX="final.bam"
```

```
TEST_BAM="${BAMDIR}/${TEST}.${BAM_SUFFIX}"
```

```
TEST_BW="${BIGWIGDIR}/${TEST}_${GENOME}_RPGC_${MARK}.bw"
```

```
if [[ "$INPUT" == "NONE" ]]; then
```

```
    HAS_INPUT=false
```

```
else
```

```
    HAS_INPUT=true
```

```
    INPUT_BAM="${BAMDIR}/${INPUT}.${BAM_SUFFIX}"
```

```
    INPUT_BW="${BIGWIGDIR}/${INPUT}_${GENOME}_RPGC_input_${MARK}.bw"
```

```
    LOG2_BW="${BIGWIGDIR}/${TEST}_vs_${INPUT}_${GENOME}_log2_RPGC_${MARK}.bw"
```

```
    SUB_BW="${BIGWIGDIR}/${TEST}_vs_${INPUT}_${GENOME}_subtract_RPGC_${MARK}.bw"
```

```
fi
```

```
# ----- Decide extendReads fragment length behavior (spatial paper parameters, but not samflag) -----
```

```
EXTEND_OPTS=()
```

```
case "$ASSAY_TYPE" in
```

```
    DRIP)
```

```
        echo "DRIP → no extension."
```

```
        EXTEND_OPTS=()
```

```
        ;;
```

```
    CUTNRUN)
```

```
        if [[ "$READ_MODE" == "PE" ]]; then
```

```
            echo "CUT&RUN PE → --extendReads"
```

```
            EXTEND_OPTS=(--extendReads)
```

```
        else
```

```
            EXTEND_OPTS=()
```

```
        fi
```

```
        ;;
```

```
    CHIP)
```

```
        if [[ "$READ_MODE" == "SE" ]]; then
```

```

        echo "CHIP SE → --extendReads 200"
        EXTEND_OPTS=(--extendReads 200)
    else
        echo "CHIP PE → no forced read filtering; use natural fragment spans"
        EXTEND_OPTS=()
    fi
;;
esac

```

```

# ----- bamCoverage: DRIP (no inputs, no extension) -----
if [[ "$ASSAY_TYPE" == "DRIP" ]]; then
    OUT="${BIGWIGDIR}/${TEST}_${GENOME}_RPGC_DRIP.bw"

    bamCoverage \
        -b "$TEST_BAM" \
        --normalizeUsing CPM \
        --binSize "$BINSIZE" \
        --smoothLength "30" \
        -p $SLURM_CPUS_PER_TASK \
        -o "$OUT"

    echo "END JOB: $(date)"
    exit 0
fi

```

```

# ----- bamCoverage: DRIPc (no inputs, no extension) -----
if [[ "$ASSAY_TYPE" == "DRIPc" ]]; then

    if [[ "$READ_MODE" == "SE" ]]; then
        PLUS_BAM="$BAMDIR/${TEST}.plus.bam"
        MINUS_BAM="$BAMDIR/${TEST}.minus.bam"
    else
        PLUS_BAM="$BAMDIR/${TEST}.R1.plus.bam"

```

```

        MINUS_BAM="$BAMDIR/${TEST}.R1.minus.bam"
    fi

    bamCoverage \
        -b "$PLUS_BAM" \
        --normalizeUsing CPM \
        --binSize 10 \
        --smoothLength 30 \
        -p $SLURM_CPUS_PER_TASK \
        -o "${BIGWIGDIR}/${TEST}_DRIPc_plus.bw"

    bamCoverage \
        -b "$MINUS_BAM" \
        --normalizeUsing CPM \
        --binSize 10 \
        --smoothLength 30 \
        -p $SLURM_CPUS_PER_TASK \
        -o "${BIGWIGDIR}/${TEST}_DRIPc_minus.bw"

    bigwigCompare \
        -b1 "${BIGWIGDIR}/${TEST}_DRIPc_plus.bw" \
        -b2 "${BIGWIGDIR}/${TEST}_DRIPc_minus.bw" \
        --operation add \
        --binSize 10 \
        -p $SLURM_CPUS_PER_TASK \
        -o "${BIGWIGDIR}/${TEST}_DRIPc_combined.bw"

    exit 0
fi

# ----- bamCoverage: CHIP & CNR input -----
if $HAS_INPUT; then
    bamCoverage -b "$INPUT_BAM" \
        --normalizeUsing RPGC \
            --effectiveGenomeSize "$EFFECTIVE_GENOME_SIZE" \
            --ignoreForNormalization chrX \
            --binSize "$BINSIZE" \

```

```

        --smoothLength "$SMOOTH" \
        "${EXTEND_OPTS[@]}" \
        -p $SLURM_CPUS_PER_TASK \
        -o "$INPUT_BW"
fi

# ----- bamCoverage: CHIP & CNR test -----
bamCoverage \
    -b "$TEST_BAM" \
    --normalizeUsing RPGC \
    --effectiveGenomeSize "$EFFECTIVE_GENOME_SIZE" \
    --ignoreForNormalization chrX \
    --binSize "$BINSIZE" \
    --smoothLength "$SMOOTH" \
    "${EXTEND_OPTS[@]}" \
    -p $SLURM_CPUS_PER_TASK \
    -o "$TEST_BW"

# ----- bigwigCompare (subtract & log2) -----
if $HAS_INPUT; then
#   bigwigCompare \
#       -b1 "$TEST_BW" \
#       -b2 "$INPUT_BW" \
#       --operation subtract \
#       --binSize "$BINSIZE" \
#       -p $SLURM_CPUS_PER_TASK \
#       -o "$SUB_BW"

    bigwigCompare \
        -b1 "$TEST_BW" \
        -b2 "$INPUT_BW" \
        --operation log2 \
        --binSize "$BINSIZE" \
        -p $SLURM_CPUS_PER_TASK \
        -o "$LOG2_BW"
fi

```

```
echo "END JOB: $(date)"
```

```
# OUTPUT
```

```
# ${TEST}_${GENOME}_RPGC_${MARK}.bw
# ${INPUT}_${GENOME}_RPGC_input_${MARK}.bw
# ${TEST}_vs_${INPUT}_${GENOME}_log2_RPGC_${MARK}.bw
# ${TEST}_vs_${INPUT}_${GENOME}_subtract_RPGC_${MARK}.bw
```

```
# NOTES
```

```
# --samFlagInclude 64 tells deepTools to keep only first mates of paired-
end reads (0x40 flag)
```

```
#####
#####
#####
#####
```

```
# Run bgwgcompare_bigwig.slurm
module load slurm
```

```
# Choose one of the below to run
```

```
# ----- DRIP -----
```

```
MARK="R-loop"
```

```
ASSAY_TYPE="DRIP"
```

```
FRAGLEN=0
```

```
READ_MODE="PE" # or SE
```

```
INPUT="NONE" # <-- REQUIRED FOR DRIP
```

```
for TEST in "${SRR_IDS[@]"; do
```

```
    sbatch bgwgcompare_bigwig.slurm \
```

```
        "$TEST" "$INPUT" "$BAMDIR" "$BIGWIGDIR" \
```

```
        "$ASSAY_TYPE" "$READ_MODE" "$FRAGLEN" "$MARK"
```

```
done
```

```

# ----- DRIPc -----
MARK="R-loop"
ASSAY_TYPE="DRIPc"
FRAGLEN=0
READ_MODE="SE" # or SE
INPUT="NONE" # <-- REQUIRED FOR DRIP

for TEST in "${SRR_IDS[@]}"; do
    sbatch bgwgcompare_bigwig.slurm \
        "$TEST" "$INPUT" "$BAMDIR" "$BIGWIGDIR" \
        "$ASSAY_TYPE" "$READ_MODE" "$FRAGLEN" "$MARK"
done

# When you load plus and minus strands, ensure that Ref gene pointing right has peak in
# plus strand. If not, switch plus and minus labels.
# Switch plus and minus strand labeling:
cd "$BIGWIGDIR"
for PLUS_BW in *_DRIPc_plus.bw; do
    base=${PLUS_BW%_DRIPc_plus.bw}
    MINUS_BW="${base}_DRIPc_minus.bw"
    TMP_BW="${base}_DRIPc_tmp_swap.bw"
    # safety check
    [[ -f "$MINUS_BW" ]] || { echo "Missing $MINUS_BW"; continue; }
    echo "Swapping:"
    echo " $PLUS_BW <-> $MINUS_BW"
    mv "$PLUS_BW" "$TMP_BW"
    mv "$MINUS_BW" "$PLUS_BW"
    mv "$TMP_BW" "$MINUS_BW"
done

```

```

# ----- CUT&RUN -----
ASSAY_TYPE="CUTNRUN"
READ_MODE="PE"           # CUT&RUN is always PE
FRAGLEN=0                # ignored for CUT&RUN PE
INPUT="${INPUT_IDS[0]}"

for TEST in "${SRR_IDS[@]"; do
    sbatch bgwgcompare_bigwig.slurm \
        "$TEST" "$INPUT" "$BAMDIR" "$BIGWIGDIR" \
        "$ASSAY_TYPE" "$READ_MODE" "$FRAGLEN" "$MARK"
done

```

```

# ----- ChIP-seq SE -----
MARK="____"             # or TF mark, e.g. SOX2, G4, TDP43
ASSAY_TYPE="CHIP"
READ_MODE="SE"
FRAGLEN=200              # 200 for H3K9me3, but MACS3 frag length for others
INPUT="${INPUT_IDS[0]}"

for TEST in "${SRR_IDS[@]"; do
    sbatch bgwgcompare_bigwig.slurm \
        "$TEST" "$INPUT" "$BAMDIR" "$BIGWIGDIR" \
        "$ASSAY_TYPE" "$READ_MODE" "$FRAGLEN" "$MARK"
done

```



```
# ----- ChIP-seq PE -----  
MARK="DAXX"           # or TF mark, e.g. SOX2, G4, TDP43  
ASSAY_TYPE="CHIP"  
READ_MODE="PE"  
FRAGLEN=0  
INPUT="${INPUT_IDS[0]}" # matched Input
```

```
for TEST in "${SRR_IDS[@]"; do  
    sbatch bgwgcompare_bigwig.slurm \  
        "$TEST" "$INPUT" "$BAMDIR" "$BIGWIGDIR" \  
        "$ASSAY_TYPE" "$READ_MODE" "$FRAGLEN" "$MARK"  
done
```

```
mv bgwgcompare*.out "$BIGWIGDIR"
```

```
# OUTPUT  
# ${TEST}_${GENOME}_RPGC_${MARK}.bw  
# ${INPUT}_${GENOME}_RPGC_input_${MARK}.bw  
# ${TEST}_vs_${INPUT}_${GENOME}_log2_RPGC_${MARK}.bw  
# ${TEST}_vs_${INPUT}_${GENOME}_subtract_RPGC_${MARK}.bw
```